AWS SAA-C03 Study Notes: Availability, Scalability & Load Balancers

1. Availability & Scalability

Scalability

- **Definition:** The ability of a system to handle greater loads by adapting.
- **Scalability != High Availability.** You can be scalable but not highly available (e.g., a massive single server).

High Availability (HA)

- **Definition:** Running your application in at least **2 Availability Zones (AZs)**.
- **Goal:** Survive a data center loss.
- **Formula:** Availability = (Total Time - Downtime) / Total Time.

2. Vertical vs. Horizontal Scaling

Feature	Vertical Scaling (Scale Up)	Horizontal Scaling (Scale Out)
Concept	Upgrade the machine (e.g., `t2.micro -> t2.large`).	Add *more* machines (e.g., 1 instance -> 3 instances).
Downtime	**Yes** (usually need to stop/start to upgrade hardware).	**No** (seamlessly add/remove nodes).
Limit	Hardware limit (you hit a ceiling).	Theoretically infinite.
Use Case	Databases (RDS), Legacy non-distributed apps.	Web Apps, Modern distributed systems.
Tools	Change Instance Type.	Auto Scaling Group (ASG), Load Balancer.

3. Elastic Load Balancers (ELB) Overview

ELBs distribute incoming traffic across multiple downstream targets (EC2, Containers, IP addresses) in multiple AZs.

Types of Load Balancers

1. **Application Load Balancer (ALB):** Layer 7 (HTTP/HTTPS/gRPC). Smart routing.
2. **Network Load Balancer (NLB):** Layer 4 (TCP/UDP). Ultra-high performance.
3. **Gateway Load Balancer (GLB):** Layer 3 (IP). Security appliances.

4. Application Load Balancer (ALB) - Layer 7

- **Ideal for:** Microservices, Container-based apps (Docker), Web Applications.
- **Protocols:** HTTP, HTTPS, HTTP/2, WebSocket, gRPC.

Listeners & Rules

- **One Listener -> Many Target Groups:** An ALB is "smart." A single listener (e.g., Port 443) can route traffic to different backends based on rules.
- **Routing Rules:**
  1. **Path-based:** `example.com/users` -> Target Group A, `example.com/orders` -> Target Group B.
  2. **Host-based:** `one.example.com` -> Target Group A, `two.example.com` -> Target Group B.
  3. **Query String / Headers:** Route based on `?id=123` or custom headers.

Target Groups

- Can target: **EC2 Instances**, **ECS Tasks**, **Lambda Functions**, **Private IP Addresses**.

5. Network Load Balancer (NLB) - Layer 4

- **Ideal for:** Real-time data, Gaming, Finance, TCP/UDP traffic.
- **Protocols:** TCP, UDP, TCP_UDP, TLS.
- **Performance:** Millions of requests per second with ultra-low latency.

Key Features

- **Static IP:** NLB provides **one static IP per AZ**. (ALB does *not*; ALB gives you a DNS name only).
- **Cross-Zone Load Balancing: Disabled by default** (to save cost). You must enable it manually.
- **One Listener -> One Target Group:** Generally, NLB is "dumb" (pass-through). It forwards TCP traffic to a target group. It does *not* look at paths or headers.

NLB Targeting an ALB

- **Pattern: Client -> NLB -> ALB -> EC2**.
- **Why do this? (Exam Question):**
  1. You need a **Fixed Static IP** for your application (NLB feature), but you also need **HTTP Path-Based Routing** (ALB feature).
  2. You want to use **AWS PrivateLink** (requires NLB) to expose a web app (ALB).

6. Gateway Load Balancer (GLB) - Layer 3 (Gateway)

- **Use Case:** Deploying, scaling, and managing **3rd Party Virtual Appliances** (Firewalls, IDS/IPS, Deep Packet Inspection).
- **How it works:**
  1. Traffic enters VPC.
  2. Route Table directs traffic to the **GLB Endpoint**.
  3. GLB routes traffic to the 3rd Party Appliance for inspection.
  4. Appliance inspects and sends it back.
  5. Traffic goes to the application.
- **Protocol:** Uses **GENEVE** protocol on **Port 6081**.
- **Transparent:** The source and destination IP addresses are preserved (the application sees the original client IP).

7. Key ELB Features & Mechanics

Sticky Sessions (Session Affinity)

- **Concept:** Ensures a client always connects to the *same* backend instance behind the LB.
- **ALB:** Uses **Cookies**.
  - *Application-based cookie:* Custom application cookie.
  - *Duration-based cookie:* Generated by LB (`AWSALB` cookie).
- **NLB:** Uses **Source IP Affinity** (Client IP is hashed to pick a server).

Cross-Zone Load Balancing

- **Without it:** Traffic is split evenly between *AZs*, not instances. (If AZ A has 1 instance and AZ B has 9 instances, the one in AZ A gets 50% of traffic -> Overload).
- **With it (Enabled):** Traffic is split evenly between *all instances* regardless of AZ.
- **Defaults:**
  - **ALB: Enabled** by default.
  - **NLB/GLB: Disabled** by default.

SSL/TLS Certificates & SNI

- **SSL Termination:** The Load Balancer handles the encryption/decryption (offloading CPU work from EC2).
- **SNI (Server Name Indication):**
  - **Problem:** You have one ALB listener (Port 443) but host multiple domains (`a.com` , `b.com`) with different SSL certificates.
  - **Solution:** SNI allows the client to "say" which hostname it wants during the TLS handshake. The ALB loads the **correct certificate** for that specific hostname.

Connection Draining (Deregistration Delay)

- **Scenario:** An instance is marked "Unhealthy" or is scaling in.
- **Mechanism:**
  1. ELB stops sending *new* requests to that instance.
  2. ELB waits for *existing* "in-flight" requests to complete.
  3. **Deregistration Delay:** The time it waits (default **300 seconds**).
- **State:** The instance status will show as `Draining`.
- **Use Case:** Prevents users from getting "Connection Reset" errors during deployments or scaling.

8. Advanced Exam Scenarios (Critical)

Client IP Preservation (X-Forwarded-For)

- **The Problem:** When ALB forwards traffic, the EC2 instance sees the **ALB's Private IP** as the source, not the user's IP.
- **The Solution (ALB):** The instance must read the `X-Forwarded-For` HTTP header to get the original Client IP.
  - `X-Forwarded-Port` and `X-Forwarded-Proto` provide the port and protocol.
- **NLB Note:** NLB preserves the client IP **automatically** when the target type is **Instance ID**. If the target type is **IP Address**, client IP preservation must be manually enabled.

Security Group Chaining (Security)

- **Scenario:** Ensure no one can bypass the Load Balancer and access EC2 directly.
- **Configuration:**
  1. **ALB Security Group:** Allow Inbound HTTP/HTTPS from `0.0.0.0/0` (The World).
  2. **EC2 Security Group:** Allow Inbound HTTP **ONLY** from the **ALB Security Group ID**.
- **Key:** Do not use IP ranges for the EC2 SG; always reference the ALB's SG ID.

Weighted Target Groups (Blue/Green Deployment)

- **Feature:** Assign weights to Target Groups for a single listener.
- **Use Case:** Migrating safely from Version 1 (Blue) to Version 2 (Green).
  - Set V1 to 90% traffic, V2 to 10% traffic.
  - Monitor errors on V2, then gradually shift weights (50/50 -> 0/100).

WAF (Web Application Firewall) Integration

- **ALB:** Supports WAF natively. You can attach a WAF Web ACL to block SQL injection or XSS.
- **NLB: Does NOT support WAF.** To use WAF with NLB, you must place CloudFront or an ALB in front of it.

Access Logs

- ELBs capture detailed information about requests (client IP, latency, request path, server response).
- **Storage:** Logs are stored in an **S3 Bucket**.
- **Requirement:** You must attach a Bucket Policy to S3 allowing the ELB service to write to it. They are *not* enabled by default.

Slow Start Mode

- **Concept:** When a new target joins the target group (e.g., after scaling out), the LB sends traffic to it linearly (ramp-up) rather than all at once.
- **Goal:** Prevents overwhelming a "cold" instance that needs time to warm up caches.